The Problem: Why ESG Investing Is Broken

Pain Point 1: ESG Scores Are Meaningless Numbers

A fund manager looks at Shell. MSCI says ESG score: **72**. Sustainalytics says: **42**. LSEG says: **58**. Same company. Three wildly different numbers.

Why? Because each provider uses different weights, different data, and different methodologies — and **none of them show their work.**

The fund manager has no idea which score to trust. They can't explain it to their client. They can't defend it to a regulator. They're flying blind with a £50 million mandate.

Pain Point 2: SFDR Compliance Costs a Fortune

Since 2022, every EU and UK fund must classify itself under SFDR:

- **Article 6:** No sustainability focus
- **Article 8:** Promotes environmental/social characteristics
- **Article 9:** Sustainable investment is the objective

To do this, fund managers must:

- Track 18 mandatory "Principal Adverse Impact" indicators
- Prove "Do No Significant Harm" (DNSH)
- Calculate taxonomy alignment
- Document everything for regulators

Current cost: £50,000–£200,000 per year in consultant fees.

There is **no affordable tool** that automates this. Small funds can't compete. Large funds pass the cost to investors.

Pain Point 3: The FCA Is Cracking Down

In 2024, the UK's Financial Conduct Authority introduced **anti-greenwashing rules.**

Every ESG claim must be:

- Clear
- Fair
- Not misleading
- Substantiated with evidence

The penalty: Fines, reputational damage, loss of license.

Fund managers are terrified because their current ESG data is a **black box.** They can't prove where a score came from. They can't audit it. They can't defend it when the regulator knocks.

Why Existing Solutions Fail

Table

Provider	What They Give	What's Missing
MSCI	ESG score + industry ranking	No rationale. No formula. Black box.
Sustainalytics	Controversy scores	No portfolio-level SFDR classification
Bloomberg Terminal	ESG data + analytics	£20,000/year. Still no SFDR automation. No audit trail.
LSEG/Refinitiv	Comprehensive ESG scores	Expensive. Opaque methodology.
Free tools (Yahoo, etc.)	Raw ESG numbers	No interpretation. No action. No compliance help.

The gap: Nobody connects **ESG data → SFDR compliance → actionable trades → audit-proof documentation** in one flow.

That's the hole Maxwell fills.

What Maxwell Actually Does (The Story)

Step 1: Ingest — "Tell Me What You Own"

A fund manager uploads their portfolio. Or types a single ticker.

Why this step matters: The user shouldn't have to learn a new interface. If they can use Excel, they can use Maxwell.

Step 2: Diagnose — "Here's What's Wrong"

Maxwell analyzes every holding and flags three things:

Table

Check	What It Means	Real-World Impact
SFDR Classification	Is this Article 6, 8, or 9?	Determines marketing claims, regulatory burden, investor eligibility
Principal Adverse Impacts	Which of the 18 mandatory indicators fail?	Must be disclosed to investors. Failures = regulatory risk.
Do No Significant Harm	Any severe controversies?	DNSH fail = cannot claim Article 8 or 9.

Why this step matters: Most funds don't know their own classification. They think they're Article 9 because they bought "green" stocks. Maxwell tells them the truth.

Step 3: Predict — "This Will Get Worse"

Maxwell's machine learning model predicts whether a company's ESG score will improve or deteriorate over the next 12 months.

Why this step matters: ESG downgrades precede stock price drops. If Maxwell flags Tesla's governance risk six months before MSCI downgrades it, the fund manager can sell early. That's **alpha.**

Step 4: Prescribe — "Here's Exactly What to Trade"

Maxwell suggests specific stock swaps:

- Same sector (maintains portfolio structure)
- Higher ESG score (improves sustainability)
- Low tracking error (minimal risk drift)

Example: Sell Shell (2.5%), buy Ørsted (2.5%). ESG +18. Tracking error +0.3.

Why this step matters: Most ESG tools say "your portfolio needs work." Maxwell says **"do this, this, and this."** Action beats analysis.

Step 5: Explain — "Here's Why, in Plain English"

Maxwell generates a narrative report:

"Your fund was Article 8 because Shell's North Sea drilling exceeds SFDR emissions thresholds and Barclays' historical governance failures violate the Do No Significant Harm principle. Replacing them with Ørsted and Lloyds brings you into Article 9 compliance while maintaining sector exposure."

Why this step matters: Fund managers must explain decisions to clients, boards, and regulators. Maxwell writes the first draft.

Step 6: Audit — "You Can Check Every Number"

Every weight, every threshold, every data source is documented in an open methodology.

Why this step matters: In an FCA greenwashing investigation, opacity is a death sentence. Transparency is survival. BlackRock can't show their formula. Maxwell can.

The Strategic Layers (Why This Wins)

Layer 1: Regulatory Necessity (SFDR)

SFDR isn't optional. It's not a nice-to-have. It's **the law** for every fund marketed in the EU or UK.

Maxwell is the **first free tool** that automates SFDR classification. That's not a feature — it's a **regulatory lifeline** for small funds.

Layer 2: Alpha Generation (ESG Momentum)

Predicting ESG deterioration before the market prices it in is **alpha.** It's the difference between a tool that saves money and a tool that makes money.

Maxwell does both.

Layer 3: Transparency as Competitive Advantage

The FCA's anti-greenwashing rules make opacity a liability. Maxwell's open methodology turns transparency into a **commercial weapon.**

A boutique fund can say: *"Our ESG process is fully auditable. Here's the GitHub."* A BlackRock competitor can't say that.

Why Each Technical Choice Matters

Table

Choice	Why We Made It	What Problem It Solves
Streamlit (not FastAPI)	Judges see a working app, not architecture. One file, instant deploy.	Complexity kills demos.
yfinance (not Bloomberg)	Free. No API key. Real Sustainalytics ESG data.	Cost barrier for small funds.
Random Forest (not XGBoost)	Harder to break. Explainable feature importance. Runs on CPU.	Judges ask "why this model?" You can answer.
Swap ranker (not PyPortfolioOpt)	Simple division: ESG improvement / tracking error. Instant. Understandable.	Full optimization is invisible in a 5-minute demo.
OpenAI API (not local LLM)	500ms response. Cheap. Good enough for structured text.	Local models are slow and unreliable on hackathon WiFi.
Open methodology	Every formula documented. GitHub public.	Regulatory defense + trust + differentiation.

The Competition Landscape (Where You Fit)

Table

Competitor	Their Strength	Their Weakness	Your Advantage
MSCI	Brand recognition	Black box. Expensive. No SFDR automation.	We show our math.

Competitor	Their Strength	Their Weakness	Your Advantage
Bloomberg	Data depth	£20k/year. No portfolio-level action.	We prescribe trades.
Sustainalytics	Controversy data	No forward prediction. No compliance layer.	We predict + comply.
Free Yahoo Finance	Free ESG scores	No interpretation. No action. No compliance.	We turn raw data into decisions.
Other hackathon teams	???	Probably building dashboards. Probably ignoring SFDR.	We solve a £200k/year real problem.

The Winning Narrative (Memorise This)

"Every UK fund manager is paying £200,000 a year to classify their portfolio under SFDR. They're using consultants, spreadsheets, and hope. Maxwell does it in 30 seconds. We don't just show ESG scores — we predict deterioration, auto-classify for compliance, prescribe exact trades, and explain everything in plain English. And unlike BlackRock, our methodology is open-source. In an FCA greenwashing crackdown, transparency is the only sustainable advantage."

What Your Team Needs to Believe

Table

Role	Why This Matters to You
Finance lead	You're solving a real £200k/year pain point. You understand the regulation. You own the credibility.

Role	Why This Matters to You
ML engineer	Your model predicts alpha. It's not just descriptive — it's predictive. That's rare.
Data engineer	You're building the pipeline that democratizes ESG data. Free tools, real impact.
LLM engineer	You're the translator between complex finance and human decision-making.
Frontend developer	Your demo wins the room. A working product beats a slide deck every time.
Pitch lead	You have a story no other team can tell. SFDR + transparency + open-source.

The One Question That Decides Everything

If a judge asks: **"Why should I care?"**

The wrong answer: *"Because ESG is important."*

The right answer: **"Because every fund manager in this room is either paying £200k for SFDR compliance or risking an FCA fine. We're the first team to solve both problems for free."**